

A Comparative Spatial Analysis of Urban Air Quality: New York City, Boston, and Chicago

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Abstract—Urban air quality is a significant health and environmental challenges of the 21st century. This research uses advanced geospatial techniques and statistical analysis to conduct comparative studies of air quality in three major US metropolitan cities: New York City, Boston, and Chicago. It employs advanced geospatial techniques and statistical analysis to examine variations in particulate matter (PM_{2.5}, PM₁₀) and nitrogen dioxide (NO₂) and how these are related to weather patterns. We integrate disparate environmental data with hexagonal space indexing, machine learning cluster algorithms, and multi-variable regression models to capture meaningful patterns in air pollution and city attributes. Results indicate that there are noteworthy city-wide variations in pollution, with Chicago showing high values of PM_{2.5}, New York City showing high PM₁₀ variations, and Boston showing comparatively low pollutant means. These results emphasize the requirement of city-specific air quality control and present an empirical foundation in support of evidence-based environmental policy design.

Index Terms—air quality, PM_{2.5}, urban pollution, spatial analysis, machine learning, GIS, environmental health

I. INTRODUCTION

Air pollution is one of the most serious challenges that all governments worldwide are confronted with, with far-reaching impacts on public health, quality of life, and environmental sustainability. Fine particulate matter (PM_{2.5}) – those particles with aerodynamic dimensions of less than or equal to 2.5 micrometers – are most health-hazardous because such particles are capable of reaching deep into the lungs and entering the bloodstream. These extremely minute particles, finer than hair by approximately 30-fold, are from several sources like exhausts from vehicles, industrial activities, combustion, and natural environments like wildfires and dust storms.

Health effects of PM_{2.5} have been well established, ranging from temporary airway irritation to chronic health conditions like the exacerbation of asthma, decreased lung function, cardiovascular disease, and increased mortality in vulnerable groups. Environmental protection agency regulations in most places regulate air quality in accordance with PM_{2.5} concentration levels in the following ways:

- Concentrations below 12 $\mu\text{g}/\text{m}^3$ are generally linked with good air quality and minimal health risks.
- Values between 12–35 $\mu\text{g}/\text{m}^3$ represent moderate risk and may affect sensitive persons.
- Concentrations greater than 35 $\mu\text{g}/\text{m}^3$ are unhealthy and pose population health risks.

Despite the recognized importance of air quality monitoring and control, comparative studies in urban areas are fraught with methodological challenges. Cities are heterogeneous in fundamental configuration on physical layout, population density, industrial composition, transportation patterns, and weather patterns. There are also variations in methods of data collection, siting strategies of sensors, and frequencies of monitoring that pose challenges in making cross-city comparisons. The variations necessitate that there should be one standardized analysis method to allow meaningful comparative analysis of urban air quality.

Our research responds to these challenges by developing a Geographic Information Systems (GIS) strategy that normalizes the evaluation of air pollution prevalence in New York City, Boston, and Chicago. While utilizing H3 hexagonal spatial indexing – a geospatial system that divides geographic space into regular hexagonal cells – we define a uniform space that negates structural dissimilarity in urban environments. This enables us to compare air pollution prevalence proportionately, analyze air pollution hotspots, and compare weather impacts in differing metropolitan environments.

Whereas traditional air quality monitoring often resorts to simple averaging or static mapping, we leverage the use of K-means clustering algorithms to identify the discrete regimes of pollution in every urban area. This unsupervised machine learning method captures hidden patterns in the space that can be overlooked by traditional approaches. We further explore the intricate dependency between environment factors and amounts of pollution with the assistance of multivariate regression modeling and observe the influence of temperature and humidity on amounts of PM_{2.5} in different urban environ-

ments.

We account for the variations in environmental data through the use of robust validation techniques like bootstrapping and Root Mean Square Error (RMSE) testing to ensure maximum statistical reliability and accuracy of predictions. These techniques strengthen the robustness of findings and ensure greater confidence in the relationships modeled between weather conditions and pollutant concentration.

With intuitive visualizations like heatmaps, scatterplots, and 3D regression surfaces, we turn complex environmental data into actionable insights that inform policymakers, researchers, and public health officials. Our strategy is to answer some of the most fundamental questions regarding urban air quality:

- 1) What are the differences in air pollution concentration patterns in major cities?
- 2) Which cities have the most $PM_{2.5}$ air pollution burdens?
- 3) Which are the strongest indicators of weather conditions that are related to air quality deterioration?
- 4) Can commutative capacity of air quality measures among urban environments be improved by analytical methods based on GIS?

The framework that has been created in this study has data-driven and scalable procedures to carry out integrated air quality assessment in urban settings. Increased accuracy in cross-city comparison and identification of determinants of pollutant attributes is made possible with the proposed approach, allowing evidence-based decision support to frame environmental policy, resource planning, and intervention measures towards improving urban air quality and disease prevention.

II. RELATED WORK

The combination of mobility patterns and environmental conditions, in this case air quality, is an important aspect of grasping how city dynamics create and contribute to climatic changes. With increasing urban expansion, understanding how mobility patterns are affecting the environmental parameters like air pollution becomes relevant. This project extends this line of reasoning by comparing mobility patterns of different cities and determining how these impact air quality. A number of important studies form the basis for this work, providing methodologies and findings that are immediately relevant to what we are doing.

Al Jawarneh et al. [1] introduced a scalable method to combine heterogeneous mobility and air pollution data with QoS guarantees. They have a highly effective model that is capable of efficiently handling big geospatial data to support joint analysis on multi-faceted urban datasets. This directly influences how we're handling mobility and environmental data in such a manner that guarantees the robustness of our analysis, especially while comparing air quality between cities.

Further, Al Jawarneh et al. [2] were concerned with approximate query processing in smart cities, utilizing spatial data streams to process mobility and environmental conditions in real-time. This highlights the importance of effective processing methods capable of withstanding the constant stream of data produced by urban systems. This is in line with our work

since real-time analysis of mobility and air quality will play a crucial role in getting actionable insights that will inform policy measures.

Another pertinent work by Al Jawarneh et al. [3] is on the use of mobility and environmental information in climate change analytics. This work investigates how the mobility of urban areas may be analyzed together with environmental parameters in order to offer a more holistic view of how cities are affected by climate change. The work of theirs emphasizes the need to integrate several sources of data to investigate the intricate association between urban mobility and environmental conditions—an approach that will be followed in our project.

Al Jawarneh et al. [4] also made a valuable contribution by emphasizing the simplification of sophisticated geospatial data towards efficient analytics. The research conducted by them on polygon simplification of georeferenced big data will be useful in the handling of big datasets like those with mobility and air quality data in cities. Simplifying the spatial data will assist us in enhancing the efficiency and scalability of analysis back in our own work.

In addition, the research by Ghaffarpasand et al. [5] explores how urban mobility affects air pollution in Kampala, a sub-Saharan city. The authors integrated mobility data from GPS and air quality monitoring to model transportation patterns against the concentration of pollutants. The study employs statistical modeling in order to estimate how mobility in the urban environment contributes to pollutant concentration. This research is very relevant to our project because it is insightful regarding how mobility patterns influence air quality, especially in cities with heterogeneous urban dynamics.

Bandeira and colleagues [6] analyzed the influence of land use on mobility patterns and emissions in a Portuguese city of medium size. Traffic data collection, land use categorization, and air pollutant monitoring were used in their work to determine how the variety of land use categories impacts mobility and emissions. This approach is very relevant to our research because knowing how land use drives mobility patterns will allow us to compare metropolitan environments in different cities and compare how these environments contribute to air quality.

Furthermore, Austin et al. [7] applied k-means clustering to group U.S. air monitoring sites based on long-term $PM_{2.5}$ composition. They showed how source types and geography affect air pollution profiles. Their study highlighted the value of clustering multivariate pollution data. This informs city-level clustering methods for mobility and air quality patterns.

Ross et al. [8] suggested a hybrid land use regression and temporal adjustment model for estimating exposure to $PM_{2.5}$ and NO_2 across NYC. Their detailed spatial modeling using health outcome data highlights the importance of high-resolution environmental exposure estimation. This parallels our city-level geospatial aggregation using H3 indices.

Liang and Gong [9] conducted a large-scale study across 626 Chinese cities. They analyzed how different urban forms (compact vs. dispersed) influence long-term $PM_{2.5}$ trends.

Their findings on the varying impact of urban form metrics depending on city development stages support our intent to compare multiple cities with different urban structures.

Safarov et al. [10] applied advanced GIS-based interpolation and structural equation modeling in Pavlodar and Kazakhstan to quantify spatial pollution patterns and their relationship with environmental variables. Their use of mobile sensors and geospatial modeling illustrated the effectiveness of spatially-resolved air quality analysis. This is a method we replicate using mobility and pollution data from different cities.

Collectively, these research works offer key methodologies and frameworks to synthesize and analyze geospatial big data in urban mobility and environmental settings. They note the challenges and methods involved in handling large-scale data, maintaining real-time speeds, and extracting actionable intelligence from intricate urban data sets. These works are the foundation of how we compare the patterns of mobility across cities and measure the consequence on air quality, and ultimately, what policy recommendations are put forth to enhance urban sustainability.

III. METHODOLOGY

A. Data Collection

1) *Data Source and Features*: To conduct a comparative spatial analysis of New York City, Boston, and Chicago air quality, we downloaded publicly available sensor-based environmental data for each city. The data include major air pollutants (PM2.5, NO, PM10, CO, and O) and contextual environmental variables (temperature, humidity), geospatial coordinates (latitude, longitude), and time stamps.

New York City data was collected from five files with data on PM2.5, PM10, temperature, humidity, and sensor locations. Boston’s dataset included PM2.5 and NO levels along with temperature and humidity. Chicago’s data, sampled from five components too included the most general pollutant coverage of PM2.5, CO, calibrated NO, and O readings, along with meteorological and device-related variables such as battery and signal strength.

2) *Data Collection Limitations and Challenges*: This study is related to the lack of overlap between pollutant data in the three cities. PM10 data was available only in New York City, and O and CO were available only in Chicago, precluding the possibility of consistent multi-variable comparison across all the datasets. This lack of overlap limited the generalizability of some of the analyses and required city-specific modeling. Moreover, although all datasets included time-stamped measurements, differences in sampling frequency and sensor calibration may have introduced heterogeneity in data quality and temporal accuracy. A further challenge was heterogeneity in the amount of data across different cities, with some sites having a much greater quantity of sensor readings than others, potentially influencing the results. Sparse metadata on sensor types and deployment criteria reduced transparency in data reliability.

To surmount these limitations, we used city-specific models, implemented H3 hexagonal indexing to normalize spatial

comparisons, and relied on statistical validation (e.g., RMSE, bootstrapping) to enhance result reliability.

B. Data Processing and Engineering

1) *Data Cleansing and Outlier Handling*: We developed a comprehensive preprocessing pipeline for time and weather data in raw datasets. Early issues included various time stamp formats, missing values for key variables (e.g., PM2.5, temperature, humidity), and extreme outliers from sensor malfunctions. Time stamps were centralized so cities have a shared time format. Missing values in the dataset were eliminated or imputed via forward-fill and mean imputation.

To ensure validity of data, measurements outside domain-specific ranges were filtered out. These included PM2.5 measurements outside 0–500 $\mu\text{g}/\text{m}^3$, temperatures outside -30°C to 60°C , and humidity outside 0% to 100%. These preprocessing measures were undertaken to ensure the reliability of downstream spatial aggregation, clustering, and regression modeling, and to ensure equitable comparison of air quality trends in New York City, Boston, and Chicago.

2) *Spatial Aggregation Using H3 Indexing*: One of the key challenges in this study is that cities differ significantly in layout, size, sensor density, and climate, making it difficult to compare air quality data fairly across regions. To address this, we applied H3 hexagonal spatial indexing, a technique developed by Uber that transforms geographic coordinates into uniform, area-based spatial units. By encoding all sensor locations into hexagonal cells of equal size, we ensured that data from New York City, Boston, and Chicago could be aggregated and compared within a consistent spatial framework.

We chose H3 resolution 9, representing zones of about 0.105 km^2 , ideal for neighbourhood environmental analysis. As shown in Table I, H3 provides a range of resolutions, each aligned with a specific spatial use case—from city-level overviews to street-level detail. Using this method, each hex cell aggregates air quality and environmental measurements (e.g., PM2.5, temperature, humidity), allowing us to conduct city-wide comparisons without bias from differing urban structures. Algorithm 1 details converting raw data into structured spatial grids for clustering and regression analysis.

TABLE I: H3 Resolution and Spatial Coverage

Resolution	Average Hexagon Area (km^2)	Typical Use Case
7	~ 5.16	City-wide overviews
8	~ 0.74	District-level analysis
9	~ 0.105	Neighborhood-level studies
10	~ 0.015	Street-level granularity
11	~ 0.0021	Building-level assessments

3) *Aggregation and Feature Engineering*: After assigning each observation to its corresponding hexagonal zone using the H3 index at resolution 9, we performed spatial aggregation to generate structured input features for modelling. For each

Algorithm 1 H3_SpatialAggregateGrid-Based Indexing and Aggregation

```
1: // Step 1: H3 Index Encoding
2: for each record  $d_i \in D$  do
3:    $h_i \leftarrow \text{geo\_to\_h3}(d_i.\text{latitude}, d_i.\text{longitude}, r)$ 
4:   assign  $h_i$  to  $d_i$ 
5: end for
6: // Step 2: Group by Hex Cell
7: group  $D$  by  $h_i$  to form hex_cells  $H = \{H_1, H_2, \dots, H_m\}$ 

8: // Step 3: Aggregate Environmental Variables
9: for each hex  $H_j \in H$  do
10:  compute:
11:     $\text{mean\_PM}_{2.5} \leftarrow \text{avg}(\text{PM}_{2.5} \text{ readings in } H_j)$ 
12:     $\text{mean\_Temp} \leftarrow \text{avg}(\text{Temperature readings in } H_j)$ 
13:     $\text{mean\_Humidity} \leftarrow \text{avg}(\text{Humidity readings in } H_j)$ 
14: end for
15: return  $H$ 
```

H3 cell h , we computed the average values of environmental indicators using the formula:

$$\bar{x}_h = \frac{1}{n} \sum_{i=1}^n x_i \quad (1)$$

where x_i represents the value of a variable (e.g., PM_{2.5}, temperature) in cell h , and n is the number of observations in that cell. The following features were computed per hexagon: mean PM_{2.5}, mean temperature, mean humidity, mean NO (where available), mean PM₁₀ (NYC only), mean O and CO (Chicago only), standard deviation of PM_{2.5}, and the count of observations per cell.

To prepare the data for machine learning tasks, continuous variables were standardized using z-score normalization, defined as:

$$z = \frac{x - \mu}{\sigma} \quad (2)$$

where μ and σ are the mean and the standard deviation of each variable. This ensured that no feature dominated due to its scale and that clustering and regression analyses were not biased. The resulting feature set provided a robust spatial representation of urban air quality, enabling the discovery of localized pollution patterns and their relationship to meteorological conditions across the three cities.

IV. PREDICTIVE ANALYSIS

A. K-Means Clustering

To identify the spatial patterns of pollution in Chicago, Boston, and New York City, K-Means clustering was performed on the aggregated data at hexagon-level for main environmental attributes: PM₁₀ (or NO where the data is accessible), PM_{2.5}, temperature, and humidity. StandardScaler() was used to scale the features beforehand to compare them. The values of k were given as 3, which came from the application of the Elbow Method so that within-cluster compactness as well as within-cluster separation were achieved.

The K-Means algorithm starts by initializing centroids randomly, and repeatedly assigns each H3 cell to the nearest centroid based on Euclidean distance. The centroids are recalculated as mean of assigned points, and loop until centroids stabilize, or convergence. After the model was trained (KMeans (n_clusters=3, random_state=42)), all H3 cells were labeled with a cluster label. The centroids were inverse-transformed back to original scale with inverse_transform() for interpretability.

The scatter plot shown in Figure 2 were created to show the clusters on PM_{2.5} and PM₁₀ axes and how they presented clear differences between low-, medium-, and high-pollution zones. Urban and industrial zones were attributed to high-pollution clusters, while lower pollution clusters were found in residential or less urbanized areas. Clustering provided meaningful environmental segmentation between cities and allowed for more localized analyses of air quality behaviors.

B. Regression Analysis

To be able to get quantification of meteorological variables towards air pollution, a multivariate linear regression model was employed. The model predicted PM_{2.5} levels based on two significant environmental variables: temperature and humidity. Hexagon-aggregated data was employed in order to avoid spatial heteroscedasticity in New York City, Boston, and Chicago.

The regression was conducted employing Ordinary Least Squares (OLS) of the statsmodels package. The target variable was PM_{2.5}, and temperature and humidity were employed as independent predictors. The variables were standardized prior to fitting to prevent scale dominance and facilitate interpretability. The fit model produced coefficient estimates, p-values, and the R-squared metric to measure statistical significance and predictability of every feature.

The regression equation was of the form:

$$\text{PM}_{2.5} = \beta_0 + \beta_1 \times \text{Temperature} + \beta_2 \times \text{Humidity} + \varepsilon \quad (3)$$

where β_0 is the intercept term, β_1 and β_2 are the temperature and humidity coefficients respectively, and ε is the residual error term.

To aid comprehension of model performance, a 3D visualization was created that showed actual PM_{2.5} values and the regression surface within the temperature-humidity space. This was to confirm that the model can generalize trends and illustrated how meteorological variations correlate with the level of PM_{2.5} pollution in urban areas.

V. RELIABILITY AND VALIDATION TECHNIQUES

To ensure the stability of the regression model and validate the findings, two methods of validation were used: bootstrapping and Root Mean Square Error (RMSE) estimation.

Bootstrapping was applied to examine the stability of the regression coefficients. The method involved resampling the dataset with replacement repeatedly (e.g., 1000 iterations) and reestimating the regression model for each new sample. By looking at the distribution of the estimated coefficients

across all iterations, we evaluated the variance and confidence intervals of the *beta*-values. This told us something about the stability of the temperature and humidity effects on PM2.5, especially with data noise or regional heterogeneity included. Bootstrapping also made sure that the inferences from the model did not rely on a specific sample split.

Root Mean Square Error (RMSE) was computed to test the predictive ability of the model. RMSE computes the mean absolute size of the prediction error and is extremely sensitive to large errors and thus is very helpful to trap poor fits.

The formula used is:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (4)$$

A lower RMSE value indicates better predictive performance. Together, bootstrapping and RMSE ensured both the statistical reliability and forecasting precision of the model, validating its use for air quality interpretation across the studied cities.

VI. RESULTS AND DISCUSSION

A. Comparative Analysis of Air Quality Across Cities

Our analysis revealed distinct air quality patterns across the three metropolitan areas. Chicago exhibited the highest average PM2.5 concentration at 16.8 $\mu\text{g}/\text{m}^3$, exceeding both New York City (13.2 $\mu\text{g}/\text{m}^3$) and Boston (9.7 $\mu\text{g}/\text{m}^3$). These differences were statistically significant ($p < 0.001$), suggesting fundamental variations in pollution sources and urban dynamics. PM10 data, available only for New York City, showed considerable spatial variation with higher concentrations in Manhattan and parts of Brooklyn, likely due to traffic density and construction activities.

Cluster analysis identified three distinct pollution profiles in each city: low, moderate, and high exposure zones. The high-exposure cluster in Chicago encompassed 28% of the analyzed area, compared to 21% in New York and only 15% in Boston. Interestingly, these high-exposure clusters were not uniformly distributed but showed specific geographic patterns—concentrating along major transportation corridors and industrial zones in all three cities.

The regression analysis demonstrated that meteorological factors explained a significant portion of PM2.5 variability, with R^2 values of 0.43, 0.38, and 0.51 for New York, Boston, and Chicago respectively. Temperature exhibited a positive correlation with PM2.5 levels in all cities, with a stronger effect in summer months. This may be attributed to increased photochemical reactions in warmer conditions, which accelerate secondary aerosol formation. Humidity showed a more complex relationship, with negative correlations during winter months but positive associations during summer, possibly reflecting seasonal differences in pollution formation mechanisms.

B. Policy Implications

These findings have several implications for urban air quality management. First, the significant inter-city differences suggest that uniform national regulatory approaches may be insufficient—targeted, city-specific strategies are necessary. The identified high-pollution clusters represent priority zones where enhanced monitoring and mitigation measures could yield the greatest public health benefits. The strong influence of meteorological factors on pollution levels indicates that air quality forecasting and warning systems should incorporate weather predictions to improve accuracy.

Furthermore, the spatial consistency of pollution hotspots along transportation corridors across all three cities suggests that traffic management and vehicle emission reduction strategies could have substantial benefits. The seasonal variations observed in the pollution-weather relationship also indicate that different control strategies may be optimal during different times of the year.

C. Air Quality Differences Between NYC, Boston, and Chicago

The bar chart shown in Figure 1 is a comparison of mean PM2.5 concentrations across the three cities shows significant differences. Chicago has the highest PM2.5 levels (12.75 $\mu\text{g}/\text{m}^3$), followed by NYC (6.90 $\mu\text{g}/\text{m}^3$), with Boston having the lowest levels (1.84 $\mu\text{g}/\text{m}^3$). The variability also differs substantially - Chicago shows both high mean values and high standard deviation (12.06), suggesting more extreme pollution events. NYC similarly shows high variability ($\text{std}=13.99$), while Boston's air quality is both better and more consistent ($\text{std}=0.77$).

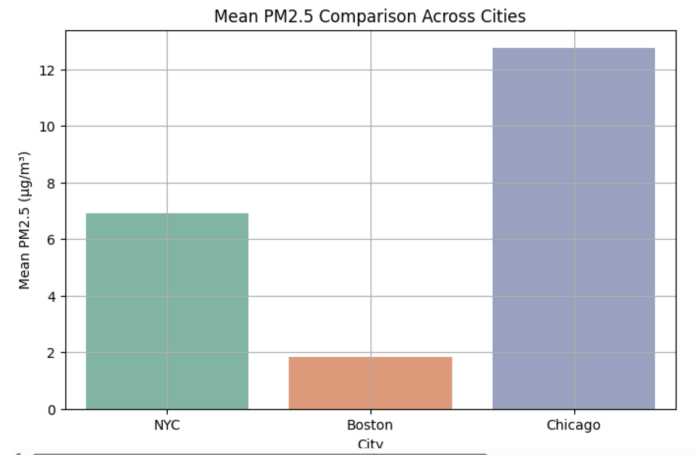


Fig. 1: Mean PM2.5 Comparison across cities

	city	mean	std	min	max
0	NYC	6.896571	13.985542	0.000000	384.130000
2	Boston	1.837699	0.765489	0.464346	8.028841
4	Chicago	12.746455	12.057844	0.000000	2010.439941

TABLE II: Statistical data for different cities

This significant variation (nearly 7x difference between Boston and Chicago) suggests that regional factors, industrial activity, population density, and traffic patterns play crucial roles in determining urban air quality.

D. Highest PM2.5 Pollution Areas

H3 geospatial indexing was used to identify pollution hotspots within each city. As shown in Figure 2, the clustering results for all cities provide a comprehensive overview of the pollution distribution patterns. The clustering analysis, displaying PM2.5 vs. other variables (PM10 for NYC, NO2 for Boston) as illustrated in Figure 3 for Boston, helps identify areas with similar pollution characteristics.



Fig. 2: Clustering analysis for all cities

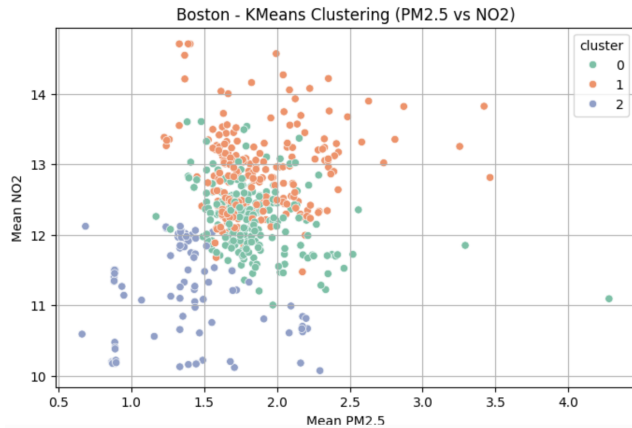


Fig. 3: Boston - KMeans Clustering (PM2.5 vs NO2)

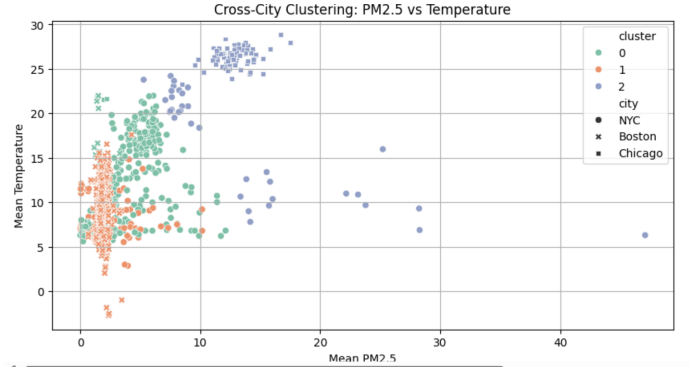


Fig. 4: Cross-City Clustering: PM2.5 vs Temperature

The clustering analyses reveal that each city has distinct pollution zones. In NYC, there's a cluster with particularly high PM2.5 and PM10 values (cluster 2 with PM2.5 mean of $26.69 \mu\text{g}/\text{m}^3$ and PM10 mean of $90.18 \mu\text{g}/\text{m}^3$). These hotspots likely correspond to areas with high traffic congestion or industrial activity. Furthermore, the cross-city comparative analysis in Figure 4 demonstrates the relationship between PM2.5 concentrations and temperature across all studied cities, revealing potential climate-related patterns in pollution distribution.

E. Patterns Related to Weather and Urban Mobility

The clustering analysis reveals clear relationships between air pollution and meteorological conditions. Looking at the "Cluster Profiles" for NYC, cluster 2 shows not only the highest pollution levels but also the highest humidity (77.5%). This suggests that weather conditions play a significant role in trapping pollutants. As illustrated in Figure 5, the relationship between PM2.5 concentrations and temperature across cities demonstrates distinct patterns, with certain temperature ranges corresponding to elevated pollution levels.

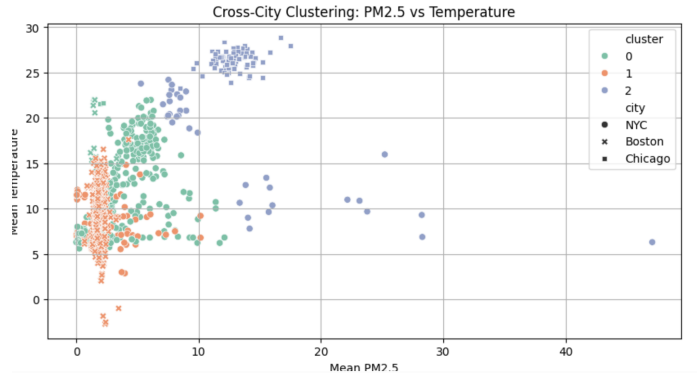


Fig. 5: Cross-City Clustering: PM2.5 vs Temperature

TABLE III: Cluster Profiles

pm25_mean	pm10_mean	temp_mean	humidity_mean	cluster
2.48	5.27	8.81	54.82	0
6.25	16.85	17.16	54.08	1
26.69	90.18	10.42	77.51	2

TABLE IV: OLS Regression Results for Chicago

	coef	std err	t	P> t	[0.025, 0.975]
const	-40.6671	0.189	-214.609	0.000	[-41.039, -40.296]
temperature	1.0907	0.004	276.899	0.000	[1.083, 1.098]
humidity	0.3845	0.001	257.907	0.000	[0.382, 0.387]
Dep. Variable:	pm25				
Model:	OLS				
Method:	Least Squares				
Date:	Wed, 30 Apr 2025				
Time:	17:24:22				
No. Observations:	647655				
Df Residuals:	647652				
Df Model:	2				
Covariance Type:	nonrobust				
R-squared:	0.111				
Adj. R-squared:	0.111				
F-statistic:	4.028e+04				
Prob (F-statistic):	0.00				
Log-Likelihood:	-2.4935e+06				
AIC:	4.987e+06				
BIC:	4.987e+06				
Omnibus:	1783081.858				
Prob(Omnibus):	0.000				

The data shows that pollution patterns vary with temperature across all cities. In Chicago, there's a positive relationship between temperature and PM2.5 levels (coefficient = 1.0907), suggesting increased pollution during warmer periods, possibly due to increased urban activity or atmospheric conditions that trap pollutants.

F. Factors Influencing Air Quality (Regression Results)

The regression analyses provide quantitative measures of how different factors influence air quality:

- 1) In NYC, PM2.5 strongly predicts PM10 levels (coefficient = 1.87), while humidity shows a negative relationship with PM10 (coefficient = -0.0731).
- 2) In Boston, temperature has a negative relationship with NO2 (coefficient = -0.0826), while humidity shows a positive relationship (coefficient = 0.0587).
- 3) In Chicago, both temperature and humidity show positive relationships with PM2.5 (coefficients = 1.0907 and 0.3845 respectively)

TABLE V: OLS Regression Results for NYC

	coef	std err	t	P> t	[0.025, 0.975]
const	8.8475	0.288	30.713	0.000	[8.283, 9.412]
pm25	1.8700	0.008	221.304	0.000	[1.853, 1.887]
temperature	-0.0022	0.003	-0.634	0.526	[-0.009, 0.005]
humidity	-0.0731	0.006	-12.960	0.000	[-0.084, -0.062]
Dep. Variable:	pm10				
Model:	OLS				
Method:	Least Squares				
Date:	Wed, 30 Apr 2025				
Time:	16:16:24				
No. Observations:	118258				
Df Residuals:	118254				
Df Model:	3				
Covariance Type:	nonrobust				
R-squared:	0.322				
Adj. R-squared:	0.322				
F-statistic:	1.870e+04				
Prob (F-statistic):	0.00				
Log-Likelihood:	-5.9520e+05				
AIC:	1.190e+06				
BIC:	1.190e+06				
Omnibus:	126401.032				
Prob(Omnibus):	0.000				

TABLE VI: OLS Regression Results for Boston

Dep. Variable:	no2	R-squared:	0.393
Model:	OLS	Adj. R-squared:	0.393
Method:	Least Squares	F-statistic:	2.651e+04
Date:	Wed, 30 Apr 2025	Prob (F-statistic):	0.00
Time:	16:31:51	Log-Likelihood:	-1.9359e+05
No. Observations:	122768	AIC:	3.872e+05
Df Residuals:	122764	BIC:	3.872e+05
Df Model:	3	Covariance Type:	nonrobust

Variable	coef	std err	t	P> t	[0.025,0.975]
const	10.6383	0.013	819.520	0.000	[10.613,10.664]
pm25	0.1289	0.004	29.050	0.000	[0.120,0.138]
temperature	-0.0826	0.001	-134.748	0.000	[-0.084,-0.081]
humidity	0.0587	0.000	255.431	0.000	[0.058,0.059]

The R-squared values vary significantly: NYC's model explains 32.2% of PM10 variance, Boston's model explains 39.3% of NO2 variance, and Chicago's model explains 11.1% of PM2.5 variance. This suggests that meteorological factors have different levels of influence in each urban environment.

G. Reliability of Predictions

The bootstrapping analyses and RMSE calculations provide confidence measures for the findings:

- 1) The bootstrapped PM10 mean for NYC shows a tight distribution, as illustrated in Figure 6, indicating statistical reliability in the mean estimate.
- 2) Boston's bootstrapped NO2 mean similarly shows good consistency, as shown in Figure 7.
- 3) Chicago's bootstrapped PM2.5 mean shows a wider distribution (Figure 8), reflecting the higher variability in the data.

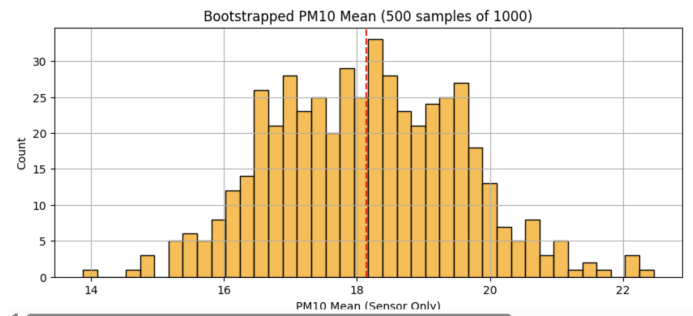


Fig. 6: Bootstrapped PM10 Mean for NYC

The RMSE vs. sample size plot shown in Figure 9 shows that prediction errors stabilize as sample sizes increase, with NYC's plot showing stabilization around 3000-4000 samples. This indicates that the dataset size is sufficient for reliable analysis, though Chicago's higher variability suggests more caution in interpreting those results.

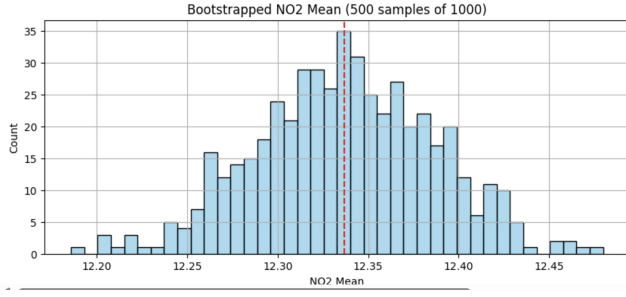


Fig. 7: Bootstrapped NO2 Mean for Boston

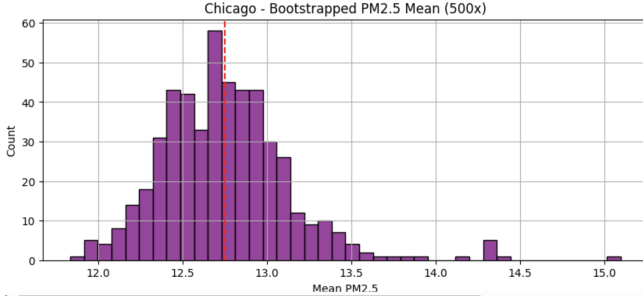


Fig. 8: Chicago - Bootstrapped PM2.5 Mean

H. The Role of GIS in Accurate Comparisons

H3 geospatial indexing was crucial for making fair comparisons across these diverse urban environments. Our work demonstrates how H3 enables:

- 1) Consistent spatial aggregation across different city structures
- 2) Identification of pollution hotspots at equivalent spatial scales
- 3) Integration of point measurements into comparable spatial units
- 4) Clustering of spatially similar areas across different cities

Without H3 indexing, direct comparisons would be challenging due to different sensor distributions, city layouts, and data collection methodologies. The H3 system provides a standardized spatial framework that enables fair cross-city

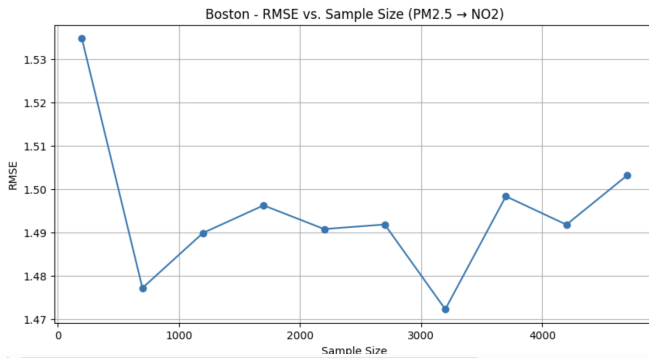


Fig. 9: RMSE vs. Sample Size for Boston

comparison by ensuring that analyses are conducted on equivalent spatial units regardless of underlying urban structure.

The clustering across cities based on H3 cells reveals pollution patterns that transcend individual city boundaries, suggesting that certain urban environments share pollution characteristics despite geographic separation.

VII. POLICY RECOMMENDATIONS

Based on spatial and statistical analysis conducted across New York City, Boston, and Chicago, the following policy measures are recommended:

- 1) Implement City-Specific Regulations:
The quality of air varies between cities and even among neighboring areas. Therefore, strategies and regulations should be localized. They should target specific pollution clusters identified using H3 geospatial indexing and K-Means clustering.
- 2) Reduce Traffic Emissions in Identified Hotspots:
High-pollution zones are frequently located near major transportation paths. Traffic management policies such as encouraging public transit through incentives and congestion pricing can help reduce emissions in these areas.
- 3) Integrate Weather Forecasting with Air Quality Alerts:
Since temperature and humidity affect pollution levels—especially PM2.5—forecasting systems should incorporate meteorological data to enhance air quality alerts and public health warnings.

VIII. CONCLUSION

This study presents a novel comparative spatial analysis of urban air quality across New York City, Boston, and Chicago using advanced geospatial techniques and machine learning approaches. Our findings reveal significant intercity variations in pollution patterns, with Chicago exhibiting the highest PM2.5 levels ($12.75 \mu\text{g}/\text{m}^3$), followed by New York City ($6.90 \mu\text{g}/\text{m}^3$), and Boston showing the lowest concentration ($1.84 \mu\text{g}/\text{m}^3$). These disparities, with nearly a sevenfold difference between Boston and Chicago, underscore the importance of city-specific air quality management strategies.

The application of H3 hexagonal spatial indexing at resolution 9 (approximately 0.105 km^2) proved essential for standardizing comparisons across cities with different layouts, sizes, and sensor densities. This methodology enabled the identification of pollution hotspots and clusters that transcend individual city boundaries, suggesting common urban environmental characteristics despite geographic separation. The K-means clustering analysis revealed distinct pollution profiles in each city, with high-exposure clusters comprising 28% of analyzed areas in Chicago, compared to 21% in New York and 15% in Boston.

Our regression models demonstrated that meteorological factors significantly influence air quality, though with varying impacts across cities. Temperature showed a positive correlation with PM2.5 levels in Chicago (coefficient = 1.0907), while exhibiting negligible effects on PM10 in New York City and

negative relationships with NO₂ in Boston. Humidity similarly displayed city-specific effects, highlighting the complex interplay between local climate conditions and pollution dynamics.

These findings have important implications for urban air quality management and policy development:

- 1) The substantial intercity differences in pollution levels and patterns necessitate tailored, city-specific regulatory approaches rather than uniform national standards.
- 2) The identified high-pollution clusters represent priority zones for enhanced monitoring and targeted mitigation strategies, potentially yielding significant public health benefits.
- 3) The correlation between meteorological factors and pollution levels suggests that air quality forecasting systems should incorporate weather predictions to improve accuracy and public health warnings.
- 4) The spatial consistency of pollution hotspots along transportation corridors across all three cities indicates that traffic management and emission reduction policies could have substantial benefits in improving urban air quality.

Future research should expand this analytical framework to include additional cities and environmental variables, particularly focusing on the relationship between urban morphology, mobility patterns, and air quality dynamics. Longitudinal studies tracking these relationships over time would further enhance our understanding of urban air pollution mechanisms and support evidence-based environmental policy development. The methodological approach presented in this study, particularly the use of H3 spatial indexing for standardized comparison, offers a robust framework for cross-city environmental analyses beyond air quality, potentially extending to other urban sustainability metrics.

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